CURRICULUM VITAE — ALIPRIYO HOORY

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Personal details

- **Nationality:** Indian
- **Date of birth:** February 12, 1999
- **Gender:** Male
- **Marital status:** Single

Academic history (in reverse chronological order)

- M.Sc.-Ph.D. Dual Degree (Physics), Indian Institute of Technology Madras, Chennai, India, 2020–Present
- B.Sc. (Physics), Ramakrishna Mission Residential College, Kolkata, India, 2017– 2020

Research interests

- Origin and evolution of primordial perturbations
  - Computation of inflationary power spectra and higher-order correlation functions
  - Comparison of inflationary models with cosmological data
  - Constraining inflationary models using machine learning algorithms
- Generation and detection of primordial gravitational waves (GWs)
  - Spectral shape of primary GWs at high frequencies
  - Scalar-induced GWs in ultra-slow roll inflation
  - Constraining non-Gaussianities associated with primordial GWs
- Observational constraints on (p)reheating
  - Effects of (p)reheating on primordial GWs
  - Formation of primordial black holes (PBHs) from preheating instabilities
  - Mass distribution of PBHs and complementary observables

Thesis work

- **Thesis topic:** Spectral shape of inflationary perturbations
- **Thesis supervisor:** Prof. L. Sriramkumar

Cosmic inflation corresponds to a brief period of accelerated expansion of the universe during the early stages of the radiation-dominated epoch. It was originally proposed to address the shortcomings of the conventional hot big bang model, such as the horizon and flatness problems. In the modern viewpoint, the most attractive aspect of the inflationary scenario lies in its ability to provide a mechanism for the generation of the primordial perturbations. In the scenario, the primordial perturbations originate from quantum fluctuations. The scalar perturbations generated during inflation are largely responsible for the anisotropies in the cosmic microwave background (CMB) and the formation of the large-scale structure. Inflationary models involving a single, slowly rolling, canonical scalar field predict a smooth, nearly scale-invariant and featureless scalar power spectrum (PS) over large scales. Such spectra are remarkably consistent with the anisotropies observed in the CMB. However, it has been repeatedly noticed in the literature that specific features in the scalar PS over large scales can lead to a better fit to the cosmological data. One of the goals of the thesis is to construct a non-parametric, machine learning pipeline to identify features in the inflationary scalar PS that provide an improved fit to the CMB data.

During inflation, in addition to the scalar perturbations, tensor perturbations or gravitational waves (GWs) are also generated from the quantum vacuum. In slow roll inflation, over large scales, as in the case of the scalar PS, the PS of the primary GWs (PGWs) is also nearly scale-invariant. However, on small scales, specifically over wave numbers that do not leave the Hubble radius during inflation, the PS of the PGWs rises quadratically. Another aim of this thesis is to systematically examine the PS of PGWs at such high wave numbers or frequencies when they are evolved post-inflation through the epochs of (p)reheating, radiation and matter domination. It has been argued that the quadratic rise in the PS of PGWs must be truncated and that the PS may instead be suppressed exponentially on such small scales. Understanding the PS of PGWs over small scales is important because they contain observable signatures related to the energy scale and duration of inflation.

Broadly, the thesis focuses on the two above-mentioned aspects of the perturbations generated during inflation. While one part is devoted to constructing features in the scalar PS on large scales that provide a better fit to the CMB data, the other part focuses on studying the observational signatures of the end of inflation and the epoch of (p)reheating on small scales. We have included below the abstracts of the three pieces of work that have been completed thus far. (The reference numbers correspond to the papers listed under publications and preprints.)

• *Origin of suppression in the PS of PGWs at high frequencies:* The PGWs are generated in the early universe from the quantum vacuum during inflation. In the standard scenario of slow roll inflation, the PS of PGWs over large scales, which leave the Hubble radius during inflation, is nearly scale-invariant. However, over very small scales, which never leave the Hubble radius, the PS of PGWs behaves as k^2, where k denotes the wave number. In this work [1], we examine the PS of PGWs at such high wave numbers or frequencies when the PGWs are evolved post-inflation, through the epochs of radiation and matter domination. Firstly, we argue that the PS has to be regularized in order to truncate the unphysical k^2 rise at high frequencies. Assuming instantaneous transitions from inflation to the epochs of radiation and matter domination, we carry out the method of adiabatic regularization to arrive at the PS of PGWs over a wide range of frequencies. We show that the process of regularization truncates the k^2 rise and that the regularized PS of PGWs oscillates with a fixed amplitude about a vanishing mean value over small scales or, equivalently, at high frequencies. Secondly, we smooth the transition from inflation to radiation domination (to be precise, we smooth the 'effective potential' governing the equation of motion of PGWs) and examine the impact of the smoothing on the regularized PS of PGWs. With the help of a linear smoothing function, we explicitly show that the smoother transition leads to a power-law suppression in the amplitude of the oscillations (about the zero mean value) of the regularized PS of PGWs over small scales that never leave the Hubble radius during inflation. Our analysis indicates that, when transitions are involved, regularization as well as smooth transitions seem essential to ensure that the correlation functions of the PGWs in real space are well behaved. We conclude with a discussion on the different directions in which the results we have obtained need to be extended.

• *Emergence of the exponential cut-off in the PS of PGWs at high frequencies:* In standard slow roll inflation, the PS of PGWs generated from the quantum vacuum rises as k^2 over wave numbers k which never leave the Hubble radius. In fact, over such small scales, the PS exhibits a similar behavior even when it is evaluated at any time after inflation. In a recent work, we had argued that the PS of PGWs has to be regularized in order to truncate the unphysical quadratic rise at large wave numbers or, equivalently, at high frequencies. Assuming instantaneous transitions from inflation to the epochs of radiation and matter domination, we had shown that the regularized PS

oscillates with a constant amplitude about a vanishing mean over small scales during these epochs. We had also smoothed the transition (or, more precisely, the 'effective potential' governing the equation of motion of GWs) from inflation to radiation domination using a linear function and exactly evaluated the regularized PS of PGWs post inflation. In such a case, we had shown that, over small scales, while the regularized PS continues to oscillate about zero, its amplitude decreases as k^{-1}. In this work [2], using the Born approximation, we examine the behavior of the regularized PS of PGWs over small scales when they are evolved through smoother and smoother transitions from inflation to the epochs of radiation and matter domination. We explicitly illustrate that, at small scales or high frequencies, the suppression in the regularized PS of PGWs occurs more and more sharply as the transition is smoothed further and further. With the help of specific examples, we also show that, in the case of completely smooth transitions described by an infinitely differentiable 'effective potential', the regularized PS of PGWs exhibits an exponential suppression on small scales. We argue that the observation of the exponential drop in the PS of PGWs can, in principle, help us determine the energy scale as well as the time of the end of inflation. We clarify a few related issues and discuss the wider implications of the results we obtain.

• *Reconstructing inflationary features on large scales using genetic algorithm:* As is well known, inflationary models involving a single, slowly rolling, scalar field predict a smooth, nearly scale-invariant and featureless scalar PS, which is remarkably consistent with the observed anisotropies in the CMB and the distribution of the large-scale structure. However, a variety of model-dependent as well as model-independent approaches suggest that certain localized features in the scalar PS can lead to a significantly better fit to the CMB data. In this work [3], we focus on three types of such features and examine whether these features can be generated in inflationary scenarios driven by a single, canonical scalar field. We consider a slowly rolling baseline model that is described by a specific time-dependence of the first slow roll parameter and we generate the desired features in the PS through suitable modifications to the functional form of the slow roll parameter. To systematically reconstruct the desired features in the scalar PS (or, equivalently, the modifications in the behavior of the first slow roll parameter) that are consistent with the data, we implement a machine learning pipeline based on the genetic algorithm (GA). Assuming the standard values for the background ΛCDM model (arrived at for a nearly scale-invariant primordial scalar PS), we apply our method to the Planck 2018 CMB data, and show that the reconstructed features improve the fit to the observed angular power spectra by $\Delta\chi^2 \lesssim -10$. Moreover, we find that GA points to other sets of background parameters and primordial features, which lead to a similar level of improvement in the fit to the data. Such alternative sets of background parameters and scalar PS offer possible pathways to alleviate existing cosmological tensions. Our approach provides effective single-field inflationary dynamics to generate features that are supported by the data.

Publications and preprints (in reverse chronological order)

The updated list of my publications can also be found on INSPIRE at `https://inspirehep.net/authors/3089025`.

Publications

1. **A. Hoory**, J. Martin, A. Paul, and L. Sriramkumar, *Primary gravitational waves at high frequencies II: Emergence of the exponential cut-off in the power spectrum*, JCAP **09**, 030 (2026) arXiv:2605.18286 [astro-ph.CO].

2. **A. Hoory**, J. Martin, A. Paul, and L. Sriramkumar, *Primary gravitational waves at high frequencies I: Origin of suppression in the power spectrum*, JCAP **05**, 080 (2026) [*arXiv:2512.03959 [astro-ph.CO]*].

Preprints

3. **A. Hoory**, D. K. Hazra, and L. Sriramkumar, *Reconstructing inflationary features on large scales using genetic algorithm*, arXiv:2604.13547 [astro-ph.CO].

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Presentations in conferences (in reverse chronological order)

1. *Power spectrum of primary gravitational waves at high frequencies*, *Physics In-house Symposium*, Department of Physics, Indian Institute of Technology Madras, Chennai, India, October 16–17, 2026.
2. *On the power spectrum of primary gravitational waves at high frequencies*, *Summer School in cosmology*, International Centre for Theoretical Physics, Trieste, Italy, June 29–July 12, 2026.
3. *Effects of regularization and smooth transitions on primary gravitational waves*, *Early Universe from Home 2026 (Online)*, February 24–27, 2026.
4. *Reconstructing primordial features on large scales using machine learning*, *Cosmic Origins and the Search for New Physics*, Indian Institute of Technology Madras, Chennai, India, January 19–23, 2026.
5. *Effects of regularization and smooth transitions on primary gravitational waves*, *Cosmic Connections: Bridging the Early and Late Universe* The Institute of Mathematical Sciences, Chennai, India, July 28–August 1, 2025.

Seminars (in reverse chronological order)

1. *On the power spectrum of primary gravitational waves at high frequencies*, Institute of Mathematics and Physics, Louvain University, Louvain-la-Neuve, Belgium, June 26, 2026.
2. *On the power spectrum of primary gravitational waves at high frequencies*, Istituto Nazionale di Astrofisica, Osservatorio di astrofisica e scienza dello spazio, Bologna, Italy, June 23, 2026.
3. *Effects of regularization and smooth transitions on primary gravitational waves*, Institut de Ciéncies del Cosmos, Universitat de Barcelona, Barcelona, Spain, June 10, 2025.

Participation in schools, workshops and conferences (in reverse chronological order)

1. *Cosmic Connections second edition: AI/ML Across Cosmic Scales*, The Institute of Mathematical Sciences, Chennai, India, September 7–11, 2026.
2. *IAGRG School on Gravitation and Cosmology*, International Centre for Theoretical Sciences, Bengaluru, India, October 14–25, 2024.
3. *Vigyan Pathshala on 'Physics of Inflation and Reheating'*, Raman Research Institute, Bengaluru, India, July 13–25, 2024.
4. *Technical and Scientific Writing Workshop*, Indian Institute of Technology, Madras, India, November 27–December 1, 2023.
5. *Statistical Methods and Machine Learning in High Energy Physics*, International Centre for Theoretical Sciences, Bengaluru, India, August 28–September 9, 2023.
6. *IAGRG School on Gravitation and Cosmology (Online)*, May 16–22, 2022.
7. *Physics of the Early Universe (Hybrid)*, International Centre for Theoretical Sciences, Bengaluru, India, January 3–12, 2022.

Fellowships and awards (in reverse chronological order)

1. *International Immersion Experience Award* from Indian Institute of Technology Madras to visit Institut d'Astrophysique de Paris, France, during May 6–July 5, 2025.
2. *Merit-cum Means Scholarship* from Indian Institute of Technology Madras, Chennai, India, 2020–2022.
3. *Swami Vivekananda Merit Cum Means Scholarship* from Government of West Bengal, India, 2017–2020.

Computational skills

- `Python`
- MATHEMATICA
- `Fortran`
- `MontePython`
- `CLASS`

Teaching assistance (in reverse chronological order)

I served as a Teaching Assistant for the following courses taught at Indian Institute of Technology Madras.

1. Introduction to General Relativity, July–November 2022.
   Course instructor: Prof. L. Sriramkumar.
2. Advanced General Relativity, January–May, 2023.
   Course instructor: Prof. L. Sriramkumar.
3. Computational Physics using Python, January–May, 2023.
   Course instructor: Dr. Samir Choudhuri.

References

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2. **Prof. Jérôme Martin**
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3. **Prof. Dhiraj Kumar Hazra**
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